

Newcomers Invade Fault-Tolerant Field, But Tandem Has Big Lead

CUPERTINO, CA — Several new companies are challenging Tandem Computers' overwhelming dominance of the fault-tolerant computer-systems market. But even though some of the upstarts offer products that seem more advanced than the leader's, Tandem officials aren't shaking just yet.

Tandem executives have good reason to shrug off the new competition. For one thing, the total market pie is growing fast enough that Tandem can probably savor a few crumbs, or even slices, for others. And, with a six-year lead on the pretenders, Tandem isn't likely to lose significant business without a major product breakthrough that would negate its learning-curve lead.

Three Who Dare

The newcomers include Stratus Computers (Natick, MA), ~~Amstel Systems (Halem, OH)~~ and Synapse Computers (Milpitas, CA). Of the three, Stratus seems to be the biggest threat to Tandem's monopoly, having begun shipments in February. Synapse has yet to announce products, much less ship them, and isn't expected to do so for at least another month. August has delivered a few systems this year, but has positioned itself in the industrial-control segment, where transaction-oriented Tandem has not made a big push.

Peter Kastner, manager of marketing development at Stratus, claims that the company has already "taken away some orders from Tandem," selling its system, for example, to a dairy company and a shoe-store chain. While current output of four systems per month is far below what Tandem produces, Kastner argues that Stratus will become more of a heavyweight in the fault-tolerant computer industry because of design features that are superior to the leader's. He also says that Stratus will make inroads based on price, since a minimum-configuration Stratus system costs about \$110,000 less than the \$260,000 price tag he puts on the equivalent Tandem package.

Focusing On Hardware

"Tandem is a software-based system. We're different. We've done it all in hardware," Kastner explains. "Every one of the components we use is self-checking during each operating cycle."

Rather than check in each cycle, Tandem's Guardian software employs periodic review. Parallel processors concurrently operate as main units, and as backups for each other. Intermittently, each CPU informs its fellows that it is available for backup duty, and monitors whether one or more of its fellows is working.

By having processors inform each other when a given set of operations begins and ends, Guardian software also insures that manipulations are not inadvertently repeated—or left hanging in mid data-stream. In addition, Guardian's Encompass program tracks procedures and guarantees that any interrupted activity is resumed from the start.

Synapse will also rely heavily on software, according to Kathryn Dillon, a financial analyst with the Gartner Group in Stamford, CT. But instead of using a parallel-processor approach, Synapse programs will allow one processor to back up several others. This should result in substantial cost savings for many users, she notes.

While the Synapse software approach may be an improvement, incorporating check procedures in hardware is a better way to go, says Stratus spokesman Kastner. That's because users can more easily extend their fault-tolerant operations. "Absolutely no user-software is involved in making existing applications fault-tolerant," he states.

Are More Bits Better?

Kastner also claims that the Stratus approach is superior because it employs 32-bit architecture. While Tandem includes 32-bit addressability in its NonStop II system, internal data paths remain in the 16-bit mode. As Kastner sees it, Stratus has a faster computational ability and a bigger memory system thanks to 32-bit capability.

Tandem marketing director Jerry Peterson doesn't think that 32-bit architecture will make much of a difference. Most NonStop customers are transaction-oriented, and they typically process only small chunks of data in each transaction. Where they need speed is in I/O functions.

The faster speed of 32-bit codes is not as important as in, say, scientific applications, where large number-crunches are involved, Peterson states. And 32-bit addressability gives them all the memory they need—up to a billion k-bytes.

"The bottom line is transaction throughput. Our customers couldn't really care whether you have 16 bits or 32 bits" in internal architecture, he claims.

Whatever the merits of that argument, involved observers like Larry Roberts, a partner in the investment-banking and venture-capital firm of Hambrecht & Quist, expect Tandem to come out with its own 32-bit product within the next 18 months. "I believe customers will strongly desire 32-bit capability," for fault-tolerant systems, since they will want such systems to be compatible with other computer applications that benefit from the longer codes, he reasons.

As a principal in one of the venture firms that have put up \$15 million to back Stratus, and \$7.9 million to back Synapse, Roberts may be a bit prejudiced about the appeal of 32-bit machines. (Synapse president Mark Leslie had once said that his company wants to produce a 32-bit microcomputer system.) But he doesn't fault Tandem for letting upstart startups take the first plunge into the longer-code systems. "I don't think they've dragged their feet at all on this," he says. "By the time you see volume shipments of 32-bit fault-tolerant machines, don't be surprised if Tandem comes out with one of its own."

Momentum Favors Tandem

Based on projections of market growth, Tandem can probably afford some competition. Looking just at CPU hardware and peripherals specifically packaged with them in fault-tolerant systems, Strategic Business Services, a San Jose-based market-research firm, puts the industry's sales last year at about \$100 million. In a survey published last November, entitled "Survivable Systems: Pitfalls and Opportunities," Strategic projects annual hardware and peripheral sales of \$2.6 billion in just six more years. Counting software sales not included in Strato-

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Newcomers Challenge Tandem In Fault Tolerance

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gie's analysis, Dataquest forecasts a \$5 billion annual volume by 1985.

Given Tandem's sales of \$208 million last year, observers credit it with near total domination of the fault-tolerant market. If the Dataquest forecast is correct, Tandem could continue its phe-

nomonal annual growth rate (the company has averaged about 106 percent revenue increases annually for the past three years) but would see its market share drop to about 70 percent. In fact, Tandem management has set its sights on being a \$1 billion company by 1985, which would put its market share at

closer to 20 percent that year. Obviously, someone is going to pick up the slack.

Stratus and Synapse clearly hope to grab some of that unclaimed business. They may well get their share, but other would-be pretenders to Tandem's domain will find it tough to operate. "I

don't think there is going to be much more room for a lot of new startups," suggests Strategic's Denis Hoyer, director of U.S. client services.

Hoyer, who edited his firm's survey of the fault-tolerant industry, predicts that much of the big market ahead will go to traditional computer powers who add higher reliability to their products and then assure total system integrity by distributing mutual intelligence to terminals via local area networks. With LANs, the loss of a terminal does not affect the operation of the total system.

In particular, 32-bit microcomputers like Intel's iAPX 432, will be "what shortens the opportunities" for Tandem-like companies. The iAPX 432 puts into firmware such attributes as interprocessor communication, processor scheduling, and data protection.

On the other hand, fault-tolerant firmware won't negate the importance of software investment as a determinant of systems selection. Users will continue to look first to vendors who already have sold them considerable software. Unfortunately for the newcomers, this factor would seem to benefit Tandem and other established vendors.

The Establishment Edge

The Securities Industry Automation Corporation (SIAC) in New York City is a case in point. It currently operates several on-line transaction systems where fault-tolerance is a prime concern. In addition to some 70 Tandem systems, SIAC uses IBM and Univac machines to keep computer-crash at a minimum.

SIAC operates the Intermarket Trading System, which links the New York, American and five regional stock exchanges and allows traders to buy and sell 50 different stocks in the most up-to-date location. This month, the ITS linked up with the NASDAQ over-the-counter market. Later this year, the expanded ITS is supposed to accommodate another 170 stocks.

While living up to these plans will mostly be a software problem, some hardware will probably be added along the way, says John McGee, vptech services for SIAC. "We're in the process now of evaluating whether we need more equipment, and we may look at some new vendors," he reports. "But we probably will pretty much go with those vendors in-house."

SIAC's huge investment in software for its current computer arsenal means any newcomer would have to show "a pretty dramatic advantage" over existing vendors to crack the lineup, McGee concludes.

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